MIL-S-28738C(EC)
15 May 1975
SUPERSEDING
MIL-S-28738B(EC)
31 March 1972

MILITARY SPECIFICATION

SWITCHING GROUP, RF, AN/UPA-61( )

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification is for a radio frequency (RF) switching group designed to provide interrogation side lobe suppression (ISLS) capability to all shipboard identification, friend-or-foe (IFF) antenna installations (see figure 1). It shall consist of a driver assembly and an RF switch assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation, Methods of
MIL-S-901	- Shock Tests, H.I. (High-Impact), Shipboard Machinery, Equipment and Systems, Requirements for (Navy)
MIL-D-3464	- Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification
MIL-E-16400 1/	- Electronic, Communication and Navigation Equipment, Naval Ships and Shore, General Specification for
MIL-E-17555	- Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-147	- Palletized and Containerized Unit Loads, 40-inch X 48-inch Pallets, Skids, Runners, or Pallet Type Base
MIL-STD-454	- Standard General Requirements for Electronic Equipment

1/ Revision F applies

FSC 5985

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MIL-STD-781	- Reliability Tests Exponential Distribution
MIL-STD-1631	- Procedure for Selection of Electronic and Electrical Parts During Equipment Design

PUBLICATIONS

MILITARY

MIL-HDBK-216 - RF Transmission Lines and Fittings

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publications, of the issue in effect on date of invitation for bids, unless otherwise stated, form a part of this specification to the extent specified herein.

NATIONAL BUREAU OF STANDARDS (NBS)

Handbook H28 - Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 General. The equipment shall be in accordance with the provisions of MIL-E-16400 except as hereinafter specified. Screws and screw-thread devices shall be in accordance with Handbook H28 (NBS).

3.1.1 First article sample. Prior to beginning production, a sample shall be tested as specified in 4.3 (see 6.2).

3.2 Environmental conditions. The equipment shall meet the following requirements of MIL-E-16400:

(a) Temperature - Class 2 for above-deck units, except the upper limit shall be +125° centigrade (C)
- Class 2 for below-deck equipment

(b) Humidity
(c) Wind speed
(d) Icing
(e) Shock
(f) Vibration
(g) Air blast (calculations only)

Note: Windspeed, icing, and air blast apply to RF switch only.

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3.2.1 Painting. The units shall be treated and painted in accordance with the requirements of MIL-E-16400.

3.3 Voltage and frequency. The equipment shall be designed to operate from a 115 volt, 60/400 Hertz (Hz), single-phase supply.

3.4 Overload protection. Circuit breakers or fuses (with spare fuses and blown-fuse indicators) shall be used in both sides of the input power line. The minimum requirement shall be that the alternating current (AC) input be current limited.

3.5 Enclosure. The degree of enclosure shall be as follows:

(a) Above-deck units - spraytight (RF switch assembly)
(b) Below-deck units - dripproof (driver assembly) See figures 1 and 2

3.6 Compatibility. The design shall be as light, rigid, and compact as state-of-the-art design allows, and shall not change or modify the mounting provisions of the associated antenna.

3.6.1 Interchangeability. Corresponding units and replaceable assemblies and parts shall be mechanically and electrically interchangeable without modification or adjustments thereto or to other articles with which the items are used.

3.6.1.1 Electromagnetic interference emission, susceptibility, and equipment and systems self-compatibility. The electromagnetic interference emission, susceptibility, and equipment and systems self-compatibility is not applicable to this specification.

3.6.1.2 RF switch. The RF switch shall be designed to be mechanically and electrically compatible with the AS-2188 antenna, and it will be tested by the command or agency concerned to ensure complete compatibility.

3.6.1.3 RF switching group. The RF switching group shall be functionally, electrically, and mechanically interchangeable with the previously procured AN/UPA-61 RF switching group. Connectors shall be of the number and type shown in figure 3 and shall be in approximately the same location with the same function and labeling as shown in figure 4.

3.7 Reliability.

3.7.1 Quantitative reliability requirement. The specified mean-time-between-failure (MTBF), θ_o as defined by MIL-STD-781, for the RF switching group shall be 5000 hours.

3.7.2 Equipment conditioning. The equipment shall be subjected to a temperature burn-in with the environment temperature equal to or greater than 50°C ±5°C with the equipment operating in the worst case mode for a cumulative operating time of 48 hours. Failures occuring during the last 24 hours shall be repaired and the burn-in resumed with a minimum of 24 hours of equipment conditioning without further failure. Equipment monitoring during burn-in shall include as a

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minimum the functional checks to assure proper operation under the worst case operating mode. No adjustments of the equipment (for example, manual compensation for drift in VCO's), other than normal operating mode changes, shall be allowed during the burn-in.

3.8 Design requirements.

3.8.1 RF switching group. The RF switching group shall consist of:

(a) Driver assembly (3.8.3) designed for below-deck mounting
(b) RF switch assembly (3.8.4) designed for above and below deck mounting

The above items and their external connections are shown in figure 1. Possible mounting configurations are shown in figure 2.

3.8.1.1 Shock requirements. Shock requirements shall be in accordance with MIL-S-901 for grade A, class I, type A equipment.

3.8.2 System inputs. The switch assembly shall be capable of operation with the inputs specified in 3.8.2.1, 3.8.2.2, and 3.8.2.3.

3.8.2.1 ISLS trigger. The ISLS trigger shall be terminated at the input to the equipment by 75 ohms ±10 percent and shall have the following characteristics:

(a)	Amplitude	- 20 ±5 volts
(b)	Polarity	- Positive
(c)	Width	- 1.0 ±0.3 microseconds (µs)
(d)	Rise time	- 0.15 µs or less
(e)	Decay time	- 0.20 µs or less
(f)	Source impedance	- Any value from 50 ohms to 10,000 ohms
(g)	Timing	- Adjustable 1.0 to 2.0 µs from the leading edge of the P_1 RF pulse

3.8.2.2 RF pulses. The RF pulses shall have the following characteristics:

(a)	Power	- 5 kilowatt (KW) peak at 0.5 percent duty cycle
(b)	Frequency	- 1030 and 1090 megaHertz (MHz)
(c)	Width	- 0.8 ±0.1 µs and 0.50 ±0.1 µs
(d)	Rise time	- 0.10 µs or less
(e)	Decay time	- 0.15 µs or less
(f)	Impedance	- 51 ±2 ohms
(g)	Minimum separation (leading edge to leading edge)	- 1.0 ±0.1 µs

3.8.2.3 Primary power. The primary power input shall be 115 volts AC ±10 percent, 60 Hz ±5 percent, or 400 Hz ±10 percent. Transient frequency operation specified in MIL-E-16400 is not required.

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3.8.3 Driver assembly. A driver assembly shall be designed for below-deck mounting, on its base only, that shall permit complete assembly withdrawal from the front. The driver assembly shall consist of the following:

(a) Driver (3.8.3.1)
(b) Power supply (3.8.3.3)
(c) Auxiliary power supply (3.8.3.1.2(c))
(d) RF switch monitor (3.8.3.4)
(e) Diplexer (3.8.3.2)

3.8.3.1 Driver. A driver shall be supplied as part of the driver assembly to provide bias and a switching pulse to the RF switch assembly by way of the diplexers and the IFF rotary joint. The driver output also shall be available at a BNC connector for monitoring and for separate routing to the switch assembly for special installations. The output impedance shall be 51 ohms nominal.

3.8.3.1.1 Driver output. The driver shall generate the positive voltage required to maintain the RF switch in the sum condition except when triggered. The driver shall generate an output switch pulse with the following characteristics and capabilities when triggered by the trigger specified in 3.8.2.1:

(a) Amplitude - negative pulse capable of driving the RF switch to the difference condition
(b) Separation of driver and switch - 5 to 500 feet (1.52 to 152.40 meters)
(c) Switch turn-on time - 0.20 µs maximum (see 6.4.1)
(d) Switch turn-off time - 0.20 µs maximum (see 6.4.1)
(e) Delay time - 0.50 µs maximum (see 6.4.2)

3.8.3.1.2 Fail-safe. The driver shall contain sail-safe provisions for the following conditions:

(a) Driver failure. The switch shall operate in the sum mode if a failure occurs in the driver circuits. A constant sum bias shall be supplied to the switch and the sum pattern shall not be deteriorated.
(b) Diode failure. The switch shall operate in the sum mode if a PIN diode short occurs. The sum pattern shall not be deteriorated and the switch shall meet the requirements of 3.8.6(a), (b), and (c).
(c) Power supply failure. The driver shall be capable of generating an auxiliary bias to enable the sum mode of operation in case of power supply or driver failure. This shall be selectable by a front panel switch that simultaneously will enable the auxiliary power supply and disable the normal power supply. An indicator light to indicate operation of the auxiliary bias shall be provided.

3.8.3.2 Diplexer. A diplexer shall be provided as part of the driver assembly to combine the transmitter RF and the direct current (DC) bias from the driver. The diplexer shall operate with the inputs specified in 3.8.2.2 and 3.8.3.1.1.

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3.8.3.2.1 Diplexer input-output. The diplexer input and output connectors shall be as specified in 3.8.7. The voltage standing wave ratio (VSWR) shall not exceed 1.25:1 when connected to an impedance of 51 ohms.

3.8.3.3 Power supply. A power supply to provide the necessary voltages shall be included as part of the driver assembly. It shall operate from the AC input specified in 3.8.2.3.

3.8.3.3.1 Indicator. An indicator lamp shall be provided on the driver assembly to indicate the presence of primary power. It shall be labeled "POWER ON".

3.8.3.4 Monitor. The driver assembly shall include a blinking light to indicate any of the failures specified in 3.8.3.1.2(a), (b), and (c). Provisions shall be made to provide a control to ground to inhibit the interrogator P_2 RF pulse in the event of a failure.1/ A multipin (4-pin) Army-Navy (AN) type jack shall be provided for the P_2 pulse inhibit (2 lines), a ground for monitor enable, plus 1 spare connection. These ground enable signals shall also occur when input power to the switch assembly is removed.

3.8.3.4.1 Automatic reset. The driver unit shall automatically reset from a fault to a normal operation when the condition causing the fault has been removed.

3.8.3.4.2 The following failure indications shall be provided to the driver assembly monitor:

(a) Open or shorted antenna cable - monitor blinker "ON"
(b) ISLS drive input missing - monitor blinker "OFF"

3.8.3.5 Construction. The construction shall employ solid-state circuitry and repairable modules and submodules.

3.8.3.6 Cooling. The driver assembly shall not employ blowers for cooling.

3.8.3.6.1 The temperature on the external surfaces of the enclosure shall be less than 40°C when the ambient temperature in the room is 25°C, when the driver assembly is normally mounted, to permit adjustment and testing during operation.

3.3.8.7 Test points. A test point shall be incorporated to monitor the switching pulse during alignment.

3.8.4 RF switch assembly. The RF switch assembly shall be a single-stripline assembly consisting of:

(a) Solid-state RF switch
(b) RF filter
(c) Diplexer

1/ A continuous ground for monitor enable shall be activated in the event of a failure.

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The assembly shall be capable of operation with the RF and video pulses specified in 3.8.2.2 and 3.8.3.1.1, respectively. The input and output impedance shall be 51 ohms nominal.

3.8.4.1 RF switch. A solid-state RF switch shall be provided utilizing field replaceable PIN diodes. It shall be integrally packaged in the RF switch assembly and shall normally be in the sum condition. The switch shall be capable of accepting the output of the driver specified in 3.8.3.1.1 via the diplexers. A TNC jack also shall be provided on the switch assembly so that the switch may be biased by means of a separate cable. When pulsed by the driver the switch shall transfer the IFF transmission from the sum output to the difference output. The switch shall be designed to remain switched to the sum port in the event of PIN diode failure.

3.8.4.2 RF filters. RF filters to attenuate radar interference shall be provided as an integral part of the RF switch assembly. Stripline techniques shall be utilized and the filters shall be located between the RF switch and each output of the assembly. The cutoff frequency shall be above 1120 MHz, and the attenuation shall increase as rapidly as practicable over the frequency range of 1120 to 1800 MHz. The minimum attenuation characteristics shall be:

(a) At 1200 MHz - 10 decibels (dB)
(b) At 1250 MHz - 20 dB
(c) At 1500 MHz - 40 dB
(d) From 1500 to 12000 MHz - minimum average isolation shall be more than 45 dB with no value less than 30 dB at any discrete frequency

3.8.4.3 Diplexer. A diplexer to separate the RF from the bias and video control pulse shall be provided as an integral part of the RF switch assembly.

3.8.5 Dimensions and weights (maximum).

(a) Driver assembly - 17.5 inches (444.5 mm) wide (to fit inside 19-inch (482.6 mm) standard rack) x 7 inches (166.8 mm) high x 16 inches (406.4 mm) deep. Weight: 35 pounds.
(b) RF switch assembly - 20 inches (508.0 mm) wide x 9 inches (228.6 mm) high x 2.75 inches (69.8 mm) thick including 0.25 inch (6.2 mm) mounting feet at each corner. Weight: nine pounds (less cables)

3.8.6 System requirements. The switch system shall have the following characteristics at both 1030 MHz and 1090 MHz:

(a) Insertion loss - 1.5 dB maximum
(b) VSWR - 1.3:1 maximum at 1030 MHz, and 1.5:1 maximum at 1090 MHz
(c) Isolation, input to output - 25 dB minimum

3.8.7 Cables and connectors. Cables and connectors shall be in accordance with MIL-HDBK-216. Connectors, unless otherwise specified, shall be type N. Cable assemblies connecting the RF switch to the antenna shall be furnished with

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appropriate connectors made up at both ends. The antenna switch cable ends shall be provided with right angle adaptors MS-90156-27C, UG-27C/U or equivalent. Both halves of all connectors shall be furnished.

3.8.8 Mounting provisions. The RF switch assembly shall be mountable on the AS-2188 antenna, either directly or by means of an interface bracket or device. A simple mounting bracket or device shall also be provided to allow below deck mounting.

3.8.9 Items not required.

3.8.9.1 Synchros. Synchros are not required.

3.8.9.2 Time meters. Time meters are not required.

3.9 Maintainability.

3.9.1 Quantitative maintainability requirements. The RF switching group shall have a specified median corrective maintenance time (Mct) of not more than 0.75 hours at a 95 percent confidence level when repair is accomplished at the organizational level of maintenance by the replacement of lowest subassemblies and chassis-mounted components (this includes electronic, electrical, electromechanical, and mechanical parts). The equipment includes localization, isolation, disassembly, interchange, reassembly, alignment and checkout for all maintenance task.

3.10 Procedure for selection of electronic and electrical parts and material during design of military items. Prescreening of nonstandard parts approval requests shall be accomplished in accordance with MIL-STD-1631.

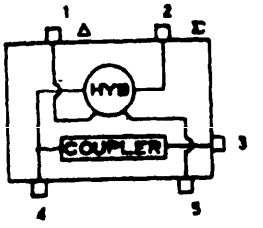
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A. HYBRID ASSEMBLY (PART OF AS-2188 ANTENNA, NOT PART OF RF SWITCH GROUP) –

1, 2 – OUTPUTS TO ANTENNA(S)

3 – TO BACKFILL RADIATOR

4, 5 – INPUTS FROM RF SWITCH
4 IS DIFF Δ, 5 IS SUM Σ

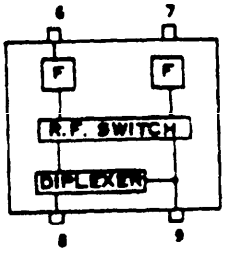


B. RF SWITCH ASSEMBLY –

6, 7 – RF SWITCH OUTPUTS

8 – RF INPUT, VIA ROTARY JOINT

9 – VIDEO CONTROL



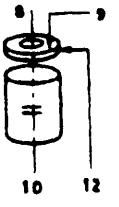
ROTARY – JOINTS (NOT PART OF EQUIP.)

FEED-THRU ~ (DC)



CAPACITIVE ~

(SLIP RING REQUIRED)



C. DRIVER ASSEMBLY –

10 – RF W/VIDEO CONTROL

11 – RF (IFF) IN AND OUT

12 – VIDEO CONTROL

13 – P_2 INHIBIT OUTPUT

14 – EXTERNAL ALARM ENABLE

15 – ISLS TRIGGER INPUT

16 – AC INPUT

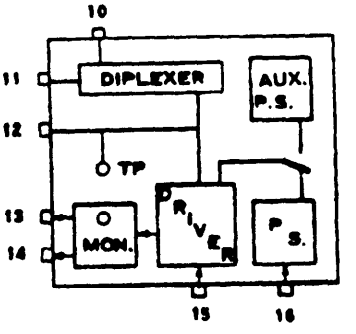


Figure 1. RF switching group.

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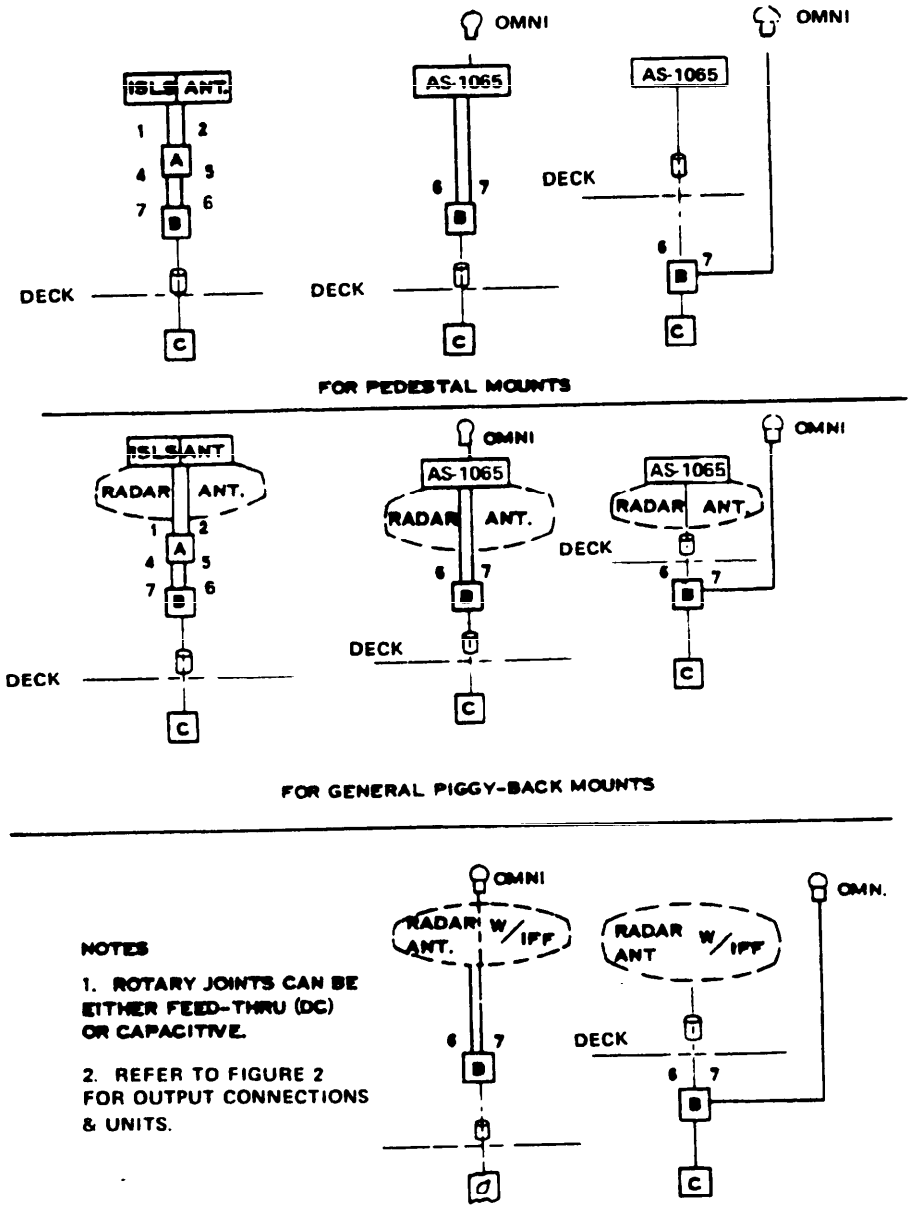
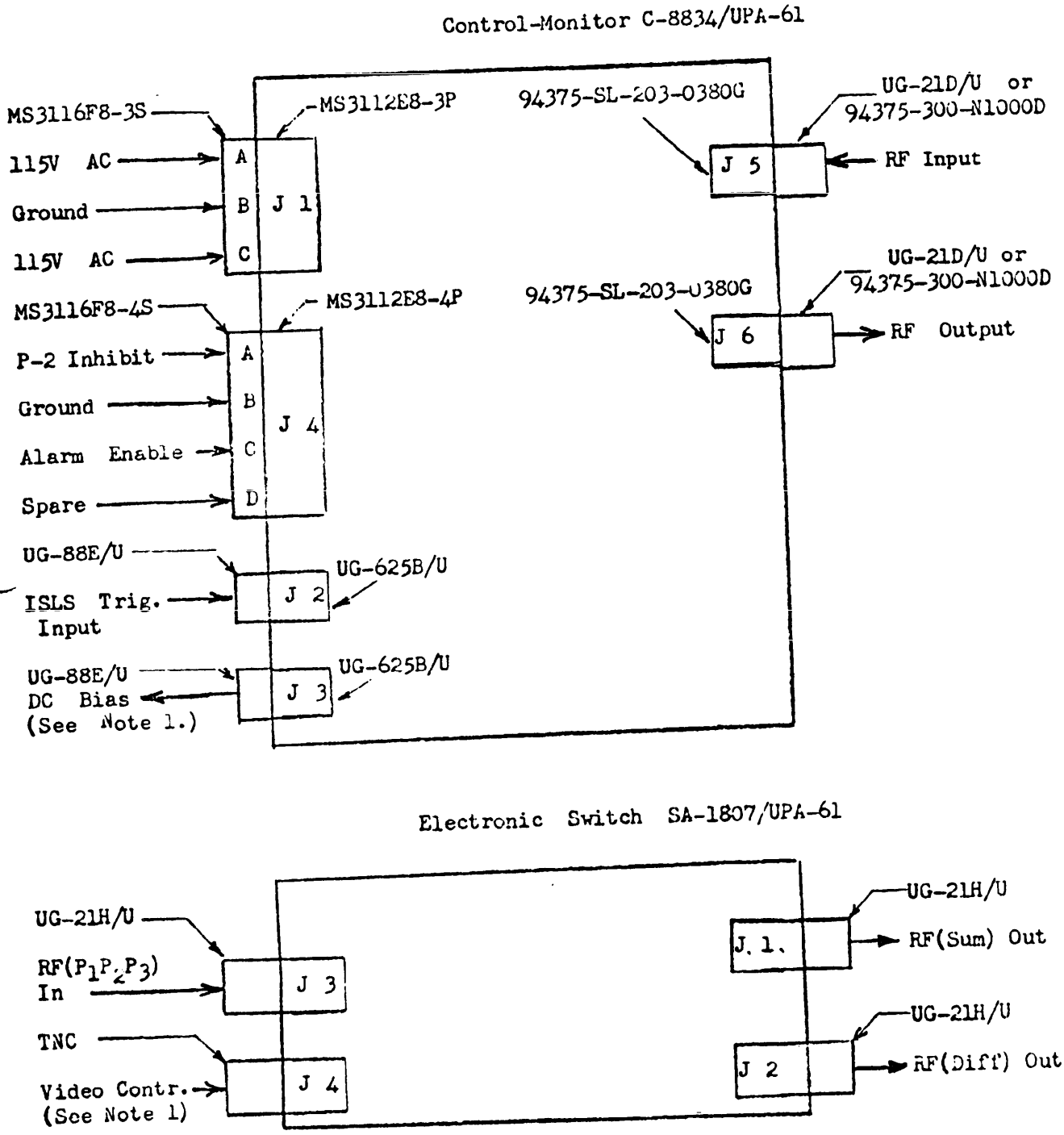


Figure 2. For mounting on radars with integral IFF.

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Note: 1. For DC Bias when required

Figure 3. AN/UPA-61 RF Switch.

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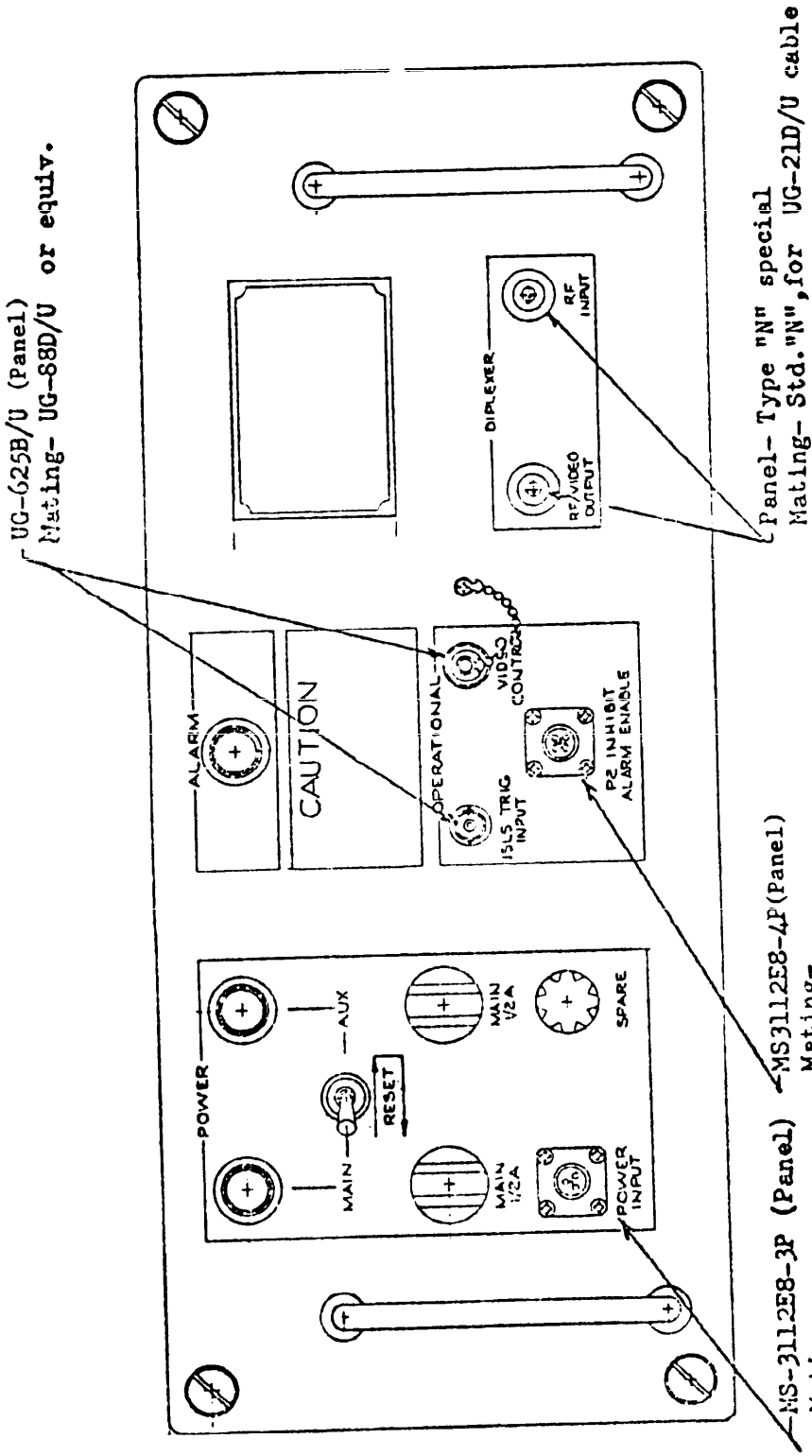


Figure 4. Front panel layout.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will include (a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied; (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered prodcut to ensure compliance with all requirements of this specification. Failure of the contractor to correct promptly deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.2 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

(1) Production inspection (see 4.4.1)
(2) Production control inspection (see 4.4.2)
(3) Environmental tests (see 4.4.3)

(c) Inspection of preparation for delivery (see 4.7)

4.3 First article inspection. Unless otherwise specified (see 6.2), two units shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in table I (see 6.3).

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Table I. First article inspection.

Examination or test	Requirement	Reference
Visual and mechanical examination	3.1	4.5.1
Operating test	3.8.2, 3.8.3.1.1, 3.8.3.1.2, 3.8.3.4, 3.8.3.4.1, 3.8.4.2, 3.8.3.1	4.5.2
Weights and dimensions	3.8.5	4.5.3
Supply line voltage and frequency	3.8.2.3	4.5.4
Heat test	3.1	4.5.5
Enclosure test	3.5	4.5.6
Temperature	3.2	4.5.7
Humidity	3.2, 3.8.6, 3.8.4.2	4.5.8
Power	3.8.2.3	4.5.9
Shock and vibration	3.2, 3.8.8, 3.8.8.1	4.5.10
Salt spray test	3.1	4.5.11
Controls and control circuits	3.8.3.1.2, 3.8.3.3.1	4.5.12
Wind speed and icing	3.1	4.5.13
Interchangeability	3.6, 3.6.1, 3.6.1.2, 3.6.13	4.5.14

4.3.1 Testing. Mounting devices shall be subjected to environmental testing with the RF switch assembly in place.

4.4 Quality conformance inspection.

4.4.1 Production inspection. Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process, such as functional and performance tests at a limited number of points, tests that detect deviations from design, tests of adjustment, and tests that detect hidden defects of material. Production inspection shall include the requirements of table II.

Table II. Production inspection.

Examination or test	Requirement	Reference
Visual and mechanical examination	3.1	4.5.1
Operating test	3.8.6, 3.8.3.21	4.5.2

4.4.2 Production control inspection. Production control inspection, including sampling, shall conform to table III and to the procedures for special inspection levels of MIL-STD-105. The inspection level shall be S-3 for normal, tightened, and reduced inspection. Production control inspection shall be performed on units that have passed production inspection. The units shall satisfactorily meet the requirements of production control inspection prior to release for shipment.

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Table III. Production control inspection

Examination or test	Requirement	Reference	AQL percent
Weights and dimensions	3.8.5	4.5.3	6.5
Supply line voltage and frequency	3.8.2.3	4.5.4	6.5
Heat test	3.1	4.5.5	6.5
Enclosure	3.5	4.5.6	6.5

4.4.2.1 Reinspection of conforming production control test units. Unless otherwise specified, sample units that have been subjected to, and have passed, production control inspections may be accepted provided they are resubjected to, and pass, production inspection after repair of all visible damage.

4.4.3 Environmental tests. Environmental tests shall be as listed in table IV and shall be performed on units that have been subjected to, and have passed, production inspection.

Table IV. Environmental tests.

Examination or test	Requirement	Reference
Temperature	3.2	4.5.7
Humidity	3.2, 3.8.6, 3.8.4.2	4.5.8
Shock and vibration	3.2, 3.8.8, 3.8.8.1	4.5.10
Salt spray test	3.1	4.5.11

4.4.3.1 Environmental test samples. Samples for the environmental inspection shall be randomly selected in accordance with the following:

(a) One sample from the first 25 production units fabricated
(b) One sample from each successive 150 units produced, or fraction thereof

4.4.3.2 Reinspection of conforming environmental sample units. Unless otherwise specified, sample units that have been subjected to, and have passed, environmental tests may be accepted provided they are resubjected to, and pass, production inspection after repair of all damage

4.4.3.3 Initial shipment. In the event the Goverment does elect to waive the requirement for first article samples, then the initial shipment of end items shall be contingent on the successful completion of the first environmental inspection (see 4.4.3).

4.5 Inspection procedures.

4.5.1 Visual and mechanical examination. Prior to performing the operating test specified by 4.5.2, each RF switching group shall be examined for workmanship in accordance with the workmanship criteria defined in MIL-STD-454 (requirement 9) to determine that the materials, parts, finish, and construction requirements specified herein have been met. Particular attention shall be given to the following:

(a) Completeness
(b) Cable runs including RF connectors
(c) Identification and marking
(d) Mounting provisions
(e) Assembly and fit
(f) Treatment for prevention of corrosion

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Upon completion of the operating test, test equipment shall be disconnected and a post-test surface examination shall be performed. Particular attention shall be given to the following:

(a) Connector or connector pin damage
(b) Cleanliness
(c) Finish
(d) Cable damage
(e) Securing of parts

4.5.2 Operating test. Test of operation of each RF switching group shall be accomplished in accordance with MIL-E-16400. As a minimum, the following characteristics shall be measured or demonstrated to verify compliance with the design requirements of this specification. During the operating test, the system inputs shall be in accordance with 3.8.2.

Characteristic	Requirement
System input (ISLS trigger and RF pulses)	3.8.2
Driver output	3.8.3.1.1
Driver fail safe operation, monitoring, and automatic resetting	3.8.3.1.2, 3.8.3.4, 3.8.3.4.1
RF filter attenuation	3.8.4.2
Impedance (driver, RF switch)	3.8.3.1, 3.8.4, 3.8.3.2.1
RF switch assembly input/output (sum and difference conditions)	3.8.4, 3.8.4.1
Switch system insertion loss, VSWR and isolation	3.8.6
Diplexer input/output	3.8.3.2.1

4.5.3 Weights and dimensions. The weights and dimensions of the driver assembly and the RF switch assembly shall be determined in accordance with MIL-E-16400 to verify compliance with the requirements of 3.8.5.

4.5.4 Supply line voltage and frequency. The RF switching group shall be subjected to supply line voltage and frequency tests in accordance with MIL-E-16400 using voltage and frequencies specified in 3.8.2.3. A transient frequency test is not required.

4.5.5 Heat test. The RF switching group shall be subjected to a heat test in accordance with MIL-E-16400, except the maximum temperature limits for the RF switch assembly shall be 125°C.

4.5.6 Enclosure test. Spray-tight (above-deck units) and dripproof (below-deck units) tests shall be performed in accordance with MIL-E-16400 to verify compliance with the requirements of 3.5. Failure of the RF switching group to operate satisfactorily or accumulation of water within the driver, or RF switch, shall be cause for rejection.

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4.5.7 Temperature. The equipment, as a system, shall be subjected to non-operating and operating condition temperature tests in accordance with MIL-E-16400, except that the upper temperature limit shall be in compliance with the requirements of 3.2. Immediately after passing through the nonoperating temperature cycle and during the operating temperature cycle, the switch system output and system characteristics requirements of 3.8.3.1.1 and 3.8.6 shall be verified.

4.5.8 Humidity test. Humidity testing shall be performed on the complete system in accordance with MIL-E-16400. Measurement of the following operating parameters shall be made prior to cycling, during cycling, and after cycling:

(a) System characteristics (3.8.6)
(b) RF filter attenuation (3.8.4.2)

4.5.9 Power. The power requirements and power factor of the RF switching group shall be measured and recorded to verify compliance with 3.8.2.3.

4.5.10 Shock and vibration. The equipment as a system shall be shock and vibration tested in accordance with MIL-E-16400 to demonstrate compliance with 3.2, 3.8.1.1, and 3.8.8. Inclination testing is not required. At the conclusion of these tests the equipment shall be subjected to, and shall meet, the requirements of the operating test (see 4.5.2).

4.5.11 Salt spray test. The salt spray test shall be made in accordance with MIL-E-16400.

4.5.12 Controls and control circuits. The equipment shall be tested as a system to:

(a) Determine the suitability of controls and control circuits for satisfactory mechanical and electrical operation
(b) Determine compliance with 3.8.3.1.2(c), 3.8.3.3.1, 3.8.3.4, 3.8.3.4.1, and requirement 28 of MIL-STD-454.

4.5.13 Wind speed and icing. The above-deck units shall be tested to demonstrate that the units will withstand the following simulated load conditions:

(a) Winds having a relative velocity of 100 knots
(b) Ice loads of 4.5 pounds per square foot

4.5.14 Inspection for interchangeability. The requirements for interchangeability of 3.6, 3.6.1, 3.6.1.2, and 3.6.1.3 shall be demonstrated on 3 systems chosen at random from each lot of 25 or less. 1/

4.6 Equipment Conditioning. All deliverable hardware (equipment and spare assemblies) shall be subjected to the temperature burn-in requirement of 3.7.2, to assure: stabilization of the required characteristics of the equipment, elimination of initial failure due to random defective components, and marginal operation due to component lot variations and/or poor assembly processes or procedures. The following cost effective approaches to capital equipment and manpower utilizatoon should be considered by the contractor:

a. The 50°C ambient temperature is a steady state condition and should be obtained by using a room or simple structure, capable of maintaining that temperature.

b. A cumulative operating time of 50°C for 48 hours on all equipments shall be required, but continuous operation is not.

c. Maximum use should be made of automatic monitoring equipment and safety devices to eliminate excess manhours.

d. A performance check for proper operation, a minimum of twice every 24 hours for equipment and just prior to enddof burn-in for equipment (and spare modules), may be performed automatically or by other empirical means approved by the contracting agency.

1/ All interchanged systems shall then be subjected to the tests of 4.5.2 and 4.5.3.

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4.7 Inspection of preparation for delivery. Inspections shall be performed to ensure conformance with Section 5.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing and marking. Unless otherwise specified, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing and marking specified in MIL-E-17555 (see 6.).

5.2 Unit packaging. Each switch shall be individually packaged in accordance with Method IIb of MIL-P-116, insuring compliance with the General paragraph under Methods of Preservation (Unit Protection) and the Physical Protection paragraph therein and shall meet the rough handling tests of MIL-P-116. Cushioning and dunnage shall be used to prevent physical or mechanical damage. Desiccant shall conform to MIL-D-3464, Type II or III.

5.3 Unitized loads. When applicable, items packed as specified in 5.1 shall be palletized or containerized in accordance with MIL-STD-147 whenever quantities to one destination equal 40 cubic feet or more. Quantities of packed items too small or impracticable for a unitized load shall be consolidated. Palletized loads shall be strapped to the pallet. Unitized loads shall not be subjected to rough handling tests. When applicable, items for shipboard end destinations shall be unitized by hull number.

5.4 Marking. In addition to any special marking required by the contract or order, each unit package and exterior container shall be marked in accordance with MIL-STD-129 (see 6.2.1). Marking shall include special marking for Method II packs.

5.5 General. Exterior containers shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent possible. Unless otherwise specified, loose-fill polystyrene shall not be used for packaging or packing. Adhesives shall not be used in the closure of wraps, bags, or boxes.

6. NOTES

6.1 Intended use. The equipment covered by this specification is intended for use in providing ISLS capability to all shipboard IFF antenna installations.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Number of first article samples to be submitted if other than specified in 4.3

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(c) Levels of preservation-packaging and packing, and applicable marking (see section 5).

6.3 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product that has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement (see 4.3).

6.4 Definitions.

6.4.1 Switch turn-on and turn-off times. Switch turn-on and turn-off times are defined as the time required for the switch to transfer 90 percent of the power from one output port to the other (see 3.8.3.1.1(c) and (d)).

6.4.2 Delay time. Delay time is defined as time elapsed from the leading edge (50 percent rise point) of the trigger pulse to the time when RF can be applied with minimum PIN diode leakage (less cable delay time) and less than 0.5 percent RF pulse width or amplitude distortion (see 3.8.3.1.1(e)).

6.5 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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